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Inferring the Detailed History of a Geothermal Field Using Sparse Data and Machine Learning

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Contents

1	Introduction	1
1.1	Measurements	1
1.2	Modelling	2
2	Literature Review	3
2.1	Belly Button Bacteria Problem (BBBP)	4
2.2	Gaussian Process Regression (GPR)	4
2.3	Ensemble Methods	5
2.4	Physics-Informed Machine Learning (PIML)	6
2.5	Mathematical Formulation	7
2.6	Research Scope	8
3	Methods	8
3.1	Gaussian Process Regression (GPR)	8
3.2	Mathematical Formulation	11
3.3	Libraries & Tools	11
3.3.1	ccandu	12
4	Results	13
4.1	Independent GPR Processes	13
4.2	GPR Processes with Conditioning	15
4.2.1	Improved Conditioning Using the ccandu Library	17
4.3	GPR with Further Relaxed Assumptions	18
4.3.1	Fitting GPR Models to Each Well	20
4.4	Using GPR on Real-Life Data	20
4.5	GPR Versus Mathematical Formulation	23
5	Discussion	24
5.1	Findings	24
5.2	Extrapolation	24
5.3	Further Work	24
6	Conclusion	25

Chapter 1

Introduction

Geothermal energy is a widely used renewable energy source that is extracted from the Earth's core [1]. Three common use cases for geothermal energy are direct-use applications, geothermal heat pumps, and electricity generation [2]. Direct-use applications for geothermal energy utilize low-temperature geothermal resources ranging between 50°C to 150°C to heat small buildings, spas, swimming pools, and greenhouses. Other common use cases involve cooking, drying fruit, timber, and vegetables, milk pasteurization, and melting snow at a large scale [2]. Geothermal heat pumps are used to heat and cool buildings. Geothermal heat pumps successfully have replaced traditional heating and cooling devices like traditional boilers, air conditioners, and furnaces due to their improved efficiency, using 25% to 50% less electricity compared to conventional heat regulation methods [2]. Also, they produce less pollution compared to traditional methods. Lastly, there is the electric power generation case, where geothermal power plants draw steam or hot liquid fluid from geothermal reservoirs, and this fluid is used to power turbines that generate electricity. Electric power generation from geothermal energy is especially common in countries such as New Zealand and Iceland, where geothermal resources, like hot springs and geysers, are abundant and accessible [1]. A geothermal resource, commonly known as a geothermal reservoir, must be present to allow for direct-use applications, geothermal heat pumps, and electricity generation. While direct-use applications and geothermal heat pumps only require the geothermal reservoir to contain hot water [3], electricity generation requires a combination of hot rocks, fluid, and permeable rocks to allow for fluid flow within the reservoir [4]. Even if these conditions occur in the geothermal reservoir, they may be unsuitable for electricity generation. Enhanced geothermal systems (EGS) deal with this by modifying the reservoir to ensure the conditions within the reservoir are adequate enough [4]. Three main geothermal power plant types are responsible for electric power generation. They are dry steam plants, flash steam plants, and binary cycle plants [1]. Dry steam plants are the oldest type of geothermal power plants and take fluid, which is mostly steam, and pushes this steam directly to a turbine. This steam drives the turbine, generating electricity. Flash steam plants take fluids greater than 182°C and pump these under high-pressure conditions into a low-pressure tank, moving the fluid from the reservoir to the surface. This large pressure change causes some of the fluid to change into vapor, which is used by the turbine to generate electricity. A second tank can exist where this process repeats itself to extract more energy. Binary cycle plants use lower-temperature fluid to generate electricity. Fluid passes through a heat exchanger with a secondary fluid. This secondary fluid has a lower boiling point than water, so when the hot water passes through it, the secondary fluid turns into steam, driving the turbine and generating electricity [4].

1.1 Measurements

Geothermal energy is often a complex system with a multitude of important measurements. Data is essential in developing viable solutions to our sparse data problem. Fortunately, we are supplied with key measurements that will enable us to apply models to our problem and,

hence, solve the problem. In particular, the mass flow, steam flow, well dryness, and status of the wells are supplied to us by the geothermal provider. Mass flow measurements for individual wells and the field itself are commonly very sparse data measurements. Contrary to this, total steam flow measurements for the geothermal field tend to be complete, but individual steam flow measurements are still sparsely measured. Well status, commonly expressed as a binary variable, is always known by the geothermal field operators. The mass flow and steam flow measurements are expressed in the standard units of kg/s while well dryness is expressed as a simple percentage. Well status is a binary variable where 1 indicates a well being on and 0 indicates a well being off. Additional measurements are also made available to us, however, are far less important for solving the problem. Brine flow, enthalpy, and net power output are all supplied to us and can be readily used if needed. They are expressed in standard units as brine flow is expressed in kg/s, enthalpy is expressed in kJ/kg, and net power output is expressed in MWe.

1.2 Modelling

Geothermal field operators need to understand how much fluid has been taken from the field and from where. Confidently knowing these two allows for informed decisions surrounding the management and future development of the field to be made. Ideally, the mass taken across time for every well in a geothermal field would be known and decisions could be readily made from this. Unfortunately, geothermal operators rarely record mass flow measurements for individual wells. Operationally, this presents significant challenges as the absence of this knowledge prevents confident decisions regarding the geothermal well to be made. Decisions relying heavily on mass flow such as whether to add fluid to the well or not could not be done, at least not with a lot of uncertainty. Modeling also becomes extremely challenging as solving relevant ODEs becomes impossible without knowing the relevant mass flow. Without being able to solve these ODEs and hence models, relevant parameters such as pressure inside the geothermal well remain unknown to geothermal operators. Developing a relevant method for imputing these missing mass flows across the geothermal wells would remove lots of uncertainty surrounding how much mass has been taken from the field and from where over time. As such, relevant operational decisions could be made with significantly reduced uncertainty and the modeling of these wells could be done as relevant ODEs could now be solved.

Chapter 2

Literature Review

The Geothermal Institute, located in Auckland, New Zealand, currently uses manual calculations to infer the mass taken from wells in a geothermal field. This process is time-consuming because it is done using an ad-hoc approach where parameters are guessed and tried. Essentially, it is a guess-and-check method. Experts in the geothermal space, such as Dr. Michael Gravatt from the Geothermal Institute at Auckland University, can often spend months determining feasible and likely combinations of these parameters so they can make the mass flow inferences they need. Not only is it time-consuming, but there is a high potential for erroneous results being produced as this method introduces lots of uncertainty, especially with such a large parameter space. As such, these current practices need to be improved as spending months to get a potentially erroneous solution with high uncertainty is costly both monetary and time-wise.

Consider figure 1 below, illustrating a simple geothermal system. There are three wells feeding into a separator that separates the steam from the steam and water mixture. The steam from these three wells feeds into a generator that produces electricity. The steam feeding into the generator is regularly recorded, but the individual wells steam and mass flow measurements are not. Suppose it was proposed to reinject additional water fluid into well B then to do this information regarding mass flow would be invaluable as it would determine things like the health of the well, how much fluid has been taken from the well, and how much fluid could safely be reinjected into the well. Consider another scenario where a new law meant that wells would need to be shut down after extracting a certain amount of fluid. This question could not be accurately answered as individual mass flow measurements for wells were not taken. Subsequently, the importance of individual well measurements, in particular mass flow, is highlighted.

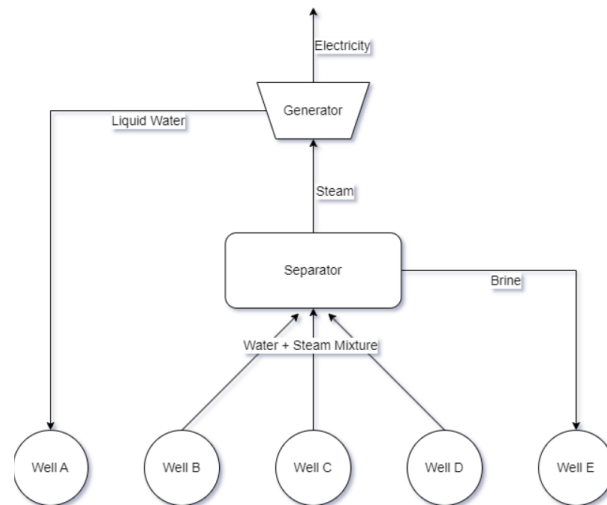


Figure 1: A simple geothermal system

These geothermal systems can become large, and thus, the lack of individual well measure-

ments, especially mass flow measurements, makes decisions regarding the geothermal field extremely challenging. Wairakei geothermal field, for example, has over 290 wells [5]. As all these wells feed into a single total steam flow production output and individual well measurements are rare, making any sort of informed decision regarding individual well operation and expansion becomes extremely challenging.

Developing a reliable, accurate, and relatively fast method for imputing these extremely sparse mass flow measurements for individual wells is extremely challenging. We will explore potential methods for solving our sparse data problem by assessing their likely effectiveness for our problem and, if relevant, supplying examples of successfully using these methods in related or unrelated contexts.

2.1 Belly Button Bacteria Problem (BBBP)

In 2011, Rob Dunn and his team of ecologists conducted a pilot study where they took swabs of the navels of 60 volunteers in order to understand what bacteria lived in this area. The pilot study resulted in the discovery of 2368 bacterial species, 1458 of which were potentially newly discovered [6]. One year later in 2012, Allen B. Downey came out with his book titled ‘Think Bayes’ [7] that applied Bayesian statistical methods to this belly button bacteria problem. His aim was to use this data to answer several related questions, such as ‘Can we estimate the prevalence of each species?’ and ‘How many additional readings are needed to raise a species’ prevalence to a threshold?’.

Downey was able to solve this problem using a Dirichlet distribution [8]. The Dirichlet distribution was able to ensure the prevalence of all species was exactly 1. For example, if there were 10 species, then if you summed the prevalence of the species, it would sum to 1. As part of the problem, he suggested generating random samples from the Dirichlet distribution using a marginal beta distribution or using n gamma distributions [9] then normalizing them by dividing through by the total.

The BBBP involves bacteria and is of an unseen species type problem. It comes from an ecological background, which is completely unrelated to geothermal energy. However, the BBBP is, in some ways, just a difficult allocation problem. The geothermal sparse data problem is also an allocation problem where the sum of steam or mass flows across the wells must sum to their relevant totals. For example, if there were 10 wells in the geothermal field then the sum of all these allocations would be exactly 1.

The Dirichlet distribution, for example, would naturally constrain allocations for wells to be between 0 and 1. Applying a marginal beta distribution and using n gamma distributions allows for random samples to be generated from this Dirichlet distribution meaning relevant confidence intervals for well allocations could be constructed. With relevant confidence intervals being constructed for well allocations across all the wells, intervals containing the likely mass flow values could be determined.

2.2 Gaussian Process Regression (GPR)

GPR is a powerful non-parametric model and stochastic process that is used in supervised machine learning problems [10]. It can be used to formulate a Bayesian framework for both regression and probabilistic classification problems where many, possibly even infinite, random variables are involved. Initially, a prior distribution is formulated where a distribution is fit across a function without considering additional information about it. This prior distribution uses the mean and covariance function to define a Gaussian process [11]. Using this allows for a function to be fit without any data observations and thus provides some levels of uncertainty before adding information to the problem. Rasmussen [10] demonstrates the prior distribution being fit using a Gaussian Process on three functions. There are no data points, and the functions are created using fitted or estimated parameters, as shown in figure 2 below.

Once the prior distribution is defined, the posterior distribution can be defined, which

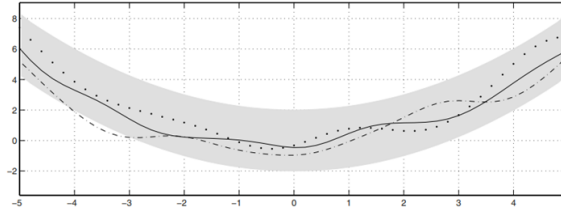


Figure 2: Example of a Gaussian Process being applied to 3 random functions creating the prior distribution. Adapted from [10]

uses both the prior distribution and existing data observations to create a more accurate function. By using a set of equations, the mean and covariance of the posterior distribution can be determined. Variance in the posterior distribution is less than the prior because it takes the prior variance and subtracts a positive term from it, which depends on the training data. In figure 3 the posterior distribution using the prior distribution in figure 2 as well as the training data is shown.

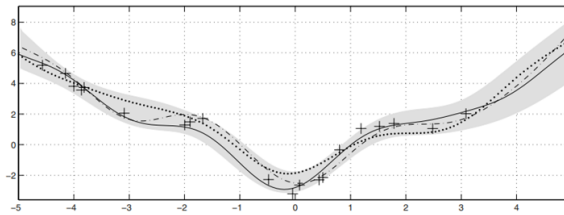


Figure 3: Example of the prior distribution and training data being used to create the posterior distribution. Adapted from [10]

As seen from figures 2 and 3, there are uncertainty intervals for the prior and posterior distribution. Using these uncertainty intervals, realizations for the target output variable can be drawn using the corresponding input variables. Multiple realizations can be performed in order to make an informed decision of the most likely value for this particular input.

While GPR hasn't been used in our exact context before, it has the potential to achieve high-quality results. Our problem introduces lots of uncertainty due to the extremely sparse data we have to work with. As such, we need a method that can first fit a model with this sparse data and second provide a range of realistic realizations under this model. Uncertainty intervals created from a GPR model would allow for mass flow realizations to be created and, hence, would naturally tackle the uncertainty issue surrounding our problem.

GPR has been applied to other unrelated problems that do deal with extremely sparse data. For example, GPR was used to predict the Young's modulus of oxide glasses using sparse datasets [12]. The research demonstrated that GPR performed well for this prediction problem and even outperformed neural networks, which are known to be very effective machine learning algorithms. Regardless of the unrelated field, the successful application of GPR for solving sparse data problems strengthens the idea of using GPR for our geothermal sparse data problem.

2.3 Ensemble Methods

Ensemble methods aim to combine multiple models in order to improve accuracy rather than just using a singular model [13]. Three common types of ensemble methods exist, with them being bagging, boosting, and stacking methods [13]. Broadly speaking, there are two distinct categories of ensemble methods with them being sequential ensemble techniques and parallel ensemble techniques. Sequential ensemble techniques build everything in

series meaning each learner is dependent on each other. Parallel-based methods, however, build everything simultaneously, which promotes independence among different learners.

Bagging methods involve three key steps, as outlined by Leo Breiman [14]. These three steps are bootstrapping, parallel training, and aggregation. The first step, bootstrapping, takes the original data and samples, with replacement, from this data until a sample of the same size as the original data is created. This process is repeated until a required number of complete samples are created. The second step, parallel training, trains each of these samples using weak learners. Finally, the third step, aggregation, uses the predictions drawn from each sample to determine the average or appropriate class, whichever one is relevant to the problem. Boosting methods iteratively combine weak classifiers to create stronger ones [15]. Various boosting methods exist, but the baseline is each iteration focuses on fixing the shortcomings of the previous iteration, so eventually a strong classifier will be established. Common boosting methods are adaptive boosting, gradient boosting, and extreme gradient boosting. Stacking methods use multiple models to create a final prediction. First, predictions are made using multiple base models [16]. Afterwards, these predictions are fed to the final model, which generates the final prediction. Figure 4 illustrates the concept of stacking concisely.

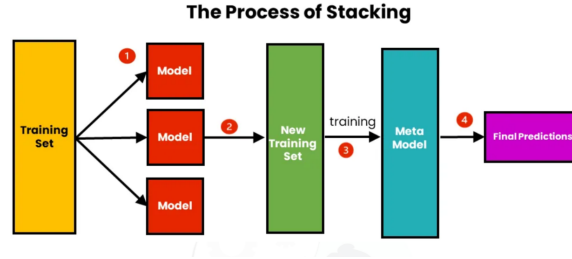


Figure 4: Illustration of stacking as an ensemble method. Adapted from [16].

These ensemble methods are highly applicable to solving this sparse data problem. By applying an appropriate ensemble method, reasonable predictions for missing mass flow could be readily made. Ensemble methods such as random forests, gradient boosting, and adaptive boosting could be used to determine likely mass flow values. The average mass flow, specified quantile range, most likely mass flow value, and many more values could be determined from these ensemble methods. As these are ensemble methods, predictions from various ensembles could deliver a robust method for predicting mass flow and thus inferring the geothermal history field.

Ensemble methods haven't been used for interpolating missing mass flow values in sparsely recorded geothermal problems. However, these ensemble methods have been used for the calibration of geothermal models before by the Geothermal Institute in Auckland [17]. Their problem focused on finding a good set of parameters to define the system accurately, known as model calibration. It was demonstrated in the conference that ensemble methods could produce accurate estimates and do so relatively efficiently [17]. These ensemble methods could be further extended to solve for missing mass flow values rather than solving a model calibration problem.

2.4 Physics-Informed Machine Learning (PIML)

Physics-informed machine learning (PIML) is a powerful tool used in machine learning focused problems. It combines machine learning techniques with physical processes to enhance the models produced [18]. Courtesy of Max Marian and Stephan Tremmel [19], a concise image displaying the relationship between data quantity and physics applied to the problem is presented.

Figure 5 demonstrates the idea that, with increased physics, the need for as much data decreases. This allows for suitable models to be created to solve sparse data problems

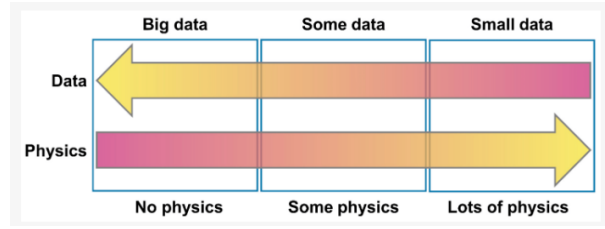


Figure 5: Relationship between data and physics. Adapted from [19]

assuming enough physics is added. While adding more physics is often appropriate, it’s important not to overconstrain anything, as this may limit the model’s ability to generate any worthwhile results.

Commonly, PIML is built using neural networks (NN) which are some of the most complex machine learning algorithms out there. The biggest strength of using NN is that they are quick to yield results due to their structure and, with good training, generate good predictions [18]. However, as with all NN they require lots of data that is broad in order to generalize well and produce good predictions [18]. In sparse data situations, this may not be possible as there isn’t enough data to produce a high-quality NN to use for PIML.

Often, when solving a geothermal problem, there is a lot of physics that needs to be carefully considered. Various things like permeability, porosity, and pressure to name a few are involved within a geothermal reservoir. As such, it can be imagined that introducing additional known values into the problem makes for more constrained and robust models and, therefore, can produce high-quality models.

PIML is a popular technique for tackling geothermal problems like reservoir management. Recently, a research paper detailing the use of PIML for reservoir management of enhanced geothermal systems (EGS) was detailed [20]. Currently, reservoir simulation is used to monitor and manage EGS which is costly and complex to do. An alternative to this approach using PIML was proposed, which would take advantage of the naturally large level of physics involved in geothermal reservoirs to manage EGS in a more cost-friendly manner. While the sparse data problem involves estimating missing mass flow values rather than reservoir management, the same principles could be applied to solve the problem. Developing a deep understanding of the geothermal field would help create a detailed model that would predict mass flow values fairly accurately, even with sparse data as the physics would compensate for this.

While this method is promising, it carries challenges surrounding the physics. Geothermal reservoirs are complex and unique, meaning each geothermal reservoir would require custom parameters to be set, which is a very challenging task for large geothermal fields. Secondly, expert domain knowledge would be needed to accurately infer and set the relevant physical parameters.

2.5 Mathematical Formulation

Assuming the problem is understood well enough and can be parameterized, a relevant mathematical model can be formed to solve directly for mass flow. This would involve solving a large system of linear equations using matrix and vector operations. Unfortunately, this hasn’t been implemented in a geothermal context so there is little literature surrounding its application for this particular sparse data problem. Furthermore, it can be imagined that solving the sparse data problem would involve using sparse matrices and vectors as well. A sparse system is able to be solved using a linear algebra approach as outlined by César Massri in his 2016 paper [21]. The paper outlines using eigenvalues, eigenvectors, and coefficient matrices to solve these sparse systems. Adapting approaches illustrated in this paper present viable ways to move forward and solve the geothermal sparse data problem.

2.6 Research Scope

This research report will focus on the application and effectiveness of using a Gaussian process regression (GPR) approach to impute missing mass flow values for individual wells using the sparse data provided. Synthetic data will be used at first to demonstrate the feasibility of a GPR approach. A proof of concept displaying this feasibility will be presented and thus real-life data supplied by Contact Energy will be used to test the GPR performance on real-life applications. It will also be investigated whether the GPR or mathematical formulation approach performs better and whether combining them is worthwhile. Creating these models will devise a robust way for missing mass flows to be predicted across wells in geothermal fields. Geothermal operators and modelers would highly benefit from such a solution as the history of a geothermal field could be inferred.

Chapter 3

Methods

Before applying Gaussian process regression (GPR) to our sparse data problem, an in-depth mathematical description of GPR is provided. This in-depth mathematical description aims to provide insight into how GPR is working and solving the problem it's presented with. Relevant libraries and tools used for solving the provided problem are also discussed. While the optimization formulation method was developed by my project partner, Daniel Clark, the relevant mathematical formulation, libraries, and tools behind it are provided, too, in an attempt to provide context to the way he solved his problem. This becomes extremely beneficial when, eventually, the GPR results are compared to the optimization formulation results.

3.1 Gaussian Process Regression (GPR)

Gaussian process regression (GPR) is an extremely clever machine learning method with complex mathematics behind it. A GPR model is made up of Gaussian processes, which are described to be a finite collection of joint Gaussian distributions defined by random variables [10]. Consider first a Gaussian distribution in the form

$$X \sim \mathcal{N}(\mu, \sigma^2) \tag{3.1.1}$$

where we are saying that X is normally distributed with a mean of μ and a variance of σ^2 where σ is the standard deviation. Naturally, this Gaussian distribution is of one-dimensional space and thus only models one random variable. As a Gaussian process requires multiple random variables, this Gaussian distribution needs to be extended somehow to include multiple random variables. A joint Gaussian distribution, commonly known as a multidimensional Gaussian distribution, depends on multiple continuous random variables [22]. The equation can be rewritten now as

$$X \sim \mathcal{N}(\mu, \Sigma) \tag{3.1.2}$$

or

$$X \sim \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_d \end{bmatrix}, \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1d} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{d1} & \sigma_{d2} & \cdots & \sigma_{dd} \end{bmatrix} \right) \quad (3.1.3)$$

where X is a normally distributed random vector, μ is a vector of means, and Σ is a covariance matrix. As such, it can be seen that the joint Gaussian distribution is defined by its mean vector and covariance matrix.

A Gaussian process takes this joint Gaussian distribution definition and extends it further by making it such that it's specified by both its mean function, m , and covariance function, k [10]. This can be expressed as

$$f \sim \text{GP}(m, k) \quad (3.1.4)$$

where we are saying that the function, f , is distributed as a Gaussian process with a mean of m and covariance of k [10]. It can be imagined that f is a distribution over a function, and hence, samples can be repeatedly drawn to produce the prior distribution [23]. This prior distribution occurs in the absence of training data and is used to gain some insight into the potential distribution of the function itself.

Incorporating data into the prior distribution allows for the creation of the posterior distribution. Consider f to be function values corresponding to the training cases and correspondingly f_* to be the unknown function values for the test cases. Then we can write out the joint distribution that contains both the test and training cases of interest to us as

$$\begin{bmatrix} \mathbf{f} \\ \mathbf{f}_* \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu} \\ \boldsymbol{\mu}_* \end{bmatrix}, \begin{bmatrix} \Sigma & \Sigma_* \\ \Sigma_*^\top & \Sigma_{**} \end{bmatrix} \right) \quad (3.1.5)$$

where μ and μ_* are the training and test means, respectively, and Σ represents the training set covariance, Σ_* represents the training-test set covariances, Σ_{**} represents the test set covariances, and $\Sigma_*^\top$ is the training-test covariance matrix transposed [10]. It's clear to us that we are interested in f_* as these values we do not know. It's possible to find an expression for $f_* | f$ using (3.1.5) by using the property

$$\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix}, \begin{bmatrix} A & C \\ C^\top & B \end{bmatrix} \right) \Bigg\} \implies \mathbf{x} | \mathbf{y} \sim \mathcal{N} (\mathbf{a} + CB^{-1}(\mathbf{y} - \mathbf{b}), A - CB^{-1}C^\top) \quad (3.1.6)$$

which essentially maps the joint distribution to a conditional expression. Applying equation (3.1.6) to equation (3.1.5) results in

$$\mathbf{f}_* | \mathbf{f} \sim \mathcal{N} (\boldsymbol{\mu}_* + \Sigma_*^\top \Sigma^{-1}(\mathbf{f} - \boldsymbol{\mu}), \Sigma_{**} - \Sigma_*^\top \Sigma^{-1} \Sigma_*) \quad (3.1.7)$$

which is the posterior distribution expression for the test cases of interest. Translating these results back to posterior processes results in the two key equations

$$m_{\mathcal{D}}(x) = m(x) + \Sigma(X, x)^\top \Sigma^{-1}(f - m) \quad (3.1.8)$$

$$k_{\mathcal{D}}(x, x') = k(x, x') - \Sigma(X, x)^\top \Sigma^{-1} \Sigma(X, x') \quad (3.1.9)$$

which represent the posterior mean, $m_{\mathcal{D}}$, and the posterior covariance matrix, $k_{\mathcal{D}}$. In this case, x is just an input to the relevant function. As the posterior mean and covariance

matrix are defined, the corresponding expression for the function conditioned on the posterior can be defined for a Gaussian process as

$$f \mid \mathcal{D} \sim \text{GP}(m_{\mathcal{D}}, k_{\mathcal{D}}) \quad (3.1.10)$$

where equation (3.1.10) combines elements from equations (3.1.4), (3.1.7), (3.1.8), and (3.1.9). All the pieces surrounding the equation are known, and hence, the posterior process and distribution can be used to generate predictions for the test cases.

This prior and posterior process is confusing mathematically and conceptually. To assist with understanding these concepts, figure 6 is presented, which displays the prior and posterior distributions for a function.

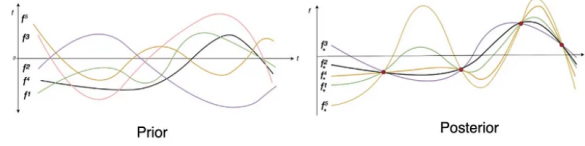


Figure 6: A simple demonstration of prior and posterior distributions. Adapted from [16].

Figure 6 demonstrates the impact conditioning can create when just a few training data points exist. The distribution becomes a lot more refined, and hence, the uncertainty and overall quality of the model improve.

Everything discussed before relies on having a good enough prior distribution to generate the posterior distribution. Often, this isn't the case where the potential behavior of the function itself isn't known at all. A prior distribution can't be fit in this case, meaning the GPR needs to be fit using the limited training data points supplied to us directly. If appropriate mean and covariance functions can be chosen for a Gaussian process, then a well-suited GPR model can be created.

The mean and covariance functions are defined by their parameters. For example, a linear equation would be defined by its slope and intercept. These parameters could be treated as hyperparameters that need to be optimized. If θ_m contains the hyperparameters for the mean function and θ_k contains the covariance function hyperparameters, then θ can be seen as the complete hyperparameter set that contains θ_m and θ_k .

The log marginal likelihood is used to determine the optimal set of hyperparameters by finding the maximum [24]. If the negative log marginal likelihood is taken, then minimizing the function will give the same optimal hyperparameter set. We will be using the negative log marginal likelihood in this report. Mathematically, the log marginal likelihood can be given as

$$L = \log p(y \mid x, \theta) = -\frac{1}{2} \log |\Sigma| - \frac{1}{2} (y - \mu)^\top \Sigma^{-1} (y - \mu) - \frac{n}{2} \log(2\pi) \quad (3.1.11)$$

where we can find the optimal set of parameters by taking the partial derivatives of the log marginal likelihood function with respect to θ_m and θ_k , respectively [10]. These partial derivatives are given as

$$\frac{\partial L}{\partial \theta_m} = -(y - \mu)^\top \Sigma^{-1} \frac{\partial \mu}{\partial \theta_m} \quad (3.1.12)$$

$$\frac{\partial L}{\partial \theta_k} = \frac{1}{2} \text{trace} \left(\Sigma^{-1} \frac{\partial \Sigma}{\partial \theta_k} \right) + \frac{1}{2} (y - \mu)^\top \frac{\partial \Sigma}{\partial \theta_k} \Sigma^{-1} \frac{\partial \Sigma}{\partial \theta_k} (y - \mu) \quad (3.1.13)$$

for the mean function and covariance function parameter sets, respectively. Equation (3.1.11) can be broken down into three terms that play different roles.

Table 3.1: Decomposition of the log marginal likelihood function into its components. Complete equation is (3.1.11)

Term	Expression
Complexity Penalty Term	$-\frac{1}{2} \log \Sigma $
Negative Quadratic Term	$-\frac{1}{2} (y - \mu)^\top \Sigma^{-1} (y - \mu)$
Log Normalization Term	$-\frac{n}{2} \log(2\pi)$

From table 3.1 it can be seen that the log marginal likelihood function is complex. The complexity penalty term acts as a measure of the model’s complexity [10]. This parameter plays a big role in ensuring the model doesn’t underfit by being too simple or doesn’t overfit but making it too complex. The negative quadratic term is the data-fit measure term that ensures the provided fit matches the data to an acceptable degree [10]. The log normalization term isn’t interesting as it has little significance [10]. Considering the first two terms discussed, it could even be seen as a tradeoff between the quality of fit and the complexity of this fit.

3.2 Mathematical Formulation

The focus of this report is not on the mathematical formulation approach to solving this sparse data problem. However, as the performance of GPR will be compared to the mathematical formulation approach later in the report, understanding the core aspects of this mathematical formulation is important and so is included in the report. The big idea of the mathematical formulation is the loss function, which is given as

$$\mathcal{L} = \sum_{t=1}^T (M_t - \sum_{i=1}^n z_{i,t} W_i(t) + \lambda \sum_{i=1}^n \sum_{t=1}^T d_{i,t} (m_{i,t} - W_i(t))^2 \quad (3.2.1)$$

where $W_i(t)$ represents the predicted mass flow rate of well i at a function of time, M_t denotes the total extraction mass flow rate recorded across all wells at time t , $m_{i,t}$ is the mass flow rate of well i at time t , $z_{i,t}$ describes whether well i was operational at time t , $d_{i,t}$ denotes whether well i ’s mass flow rate was measured at time t , T represents the total number of time points, and n is the number of wells in the geothermal field. This loss function can be minimized by differentiating the loss with respect to the wells and then setting this equation equal to zero.

3.3 Libraries & Tools

In order to develop a usable model to solve our sparse data problem, various libraries and tools were used. Scikit-Learn was used to generate the GPR models that I used to develop a solution to this problem. SymPy and research virtual machines were used to develop solutions using the optimization formulation technique.

Python was used to develop both GPR and optimization formulation techniques. Python was the language selected by both myself and my project partner, Daniel Clark because it was very familiar to us, versatile enough to perform tasks we needed it to do and has suitable IDEs that allow for advanced debugging approaches. Very common libraries used that won’t be discussed in large detail were NumPy, Matplotlib, and pandas. Numpy allowed us to easily work with arrays, a common data structure in Python. Matplotlib performed the plotting functions, such as plotting the many graphs developed during the research. The Python library pandas allowed us to easily read CSV files that contained the relevant data supplied to us.

Scikit-Learn is a machine-learning library for Python. It contains many machine-learning algorithms such as linear models, decision trees, and, of course, Gaussian processes. It's able to generate Gaussian processes for both regression and classification purposes, but our focus lies in the regression aspect. The advantages of using Scikit-Learn are that it's very accessible, easy to use, free, and well-integrated with Python. Unfortunately, Scikit-Learn is also not as powerful as other GPR-focused packages like GPy and GPflow.

SymPy is a library used for symbolic computation. It's useful when an equation needs to be typed out that relies on many parameters that need to be solved. While not useful for GPR applications, this package was extremely useful for developing the optimization formulation solution as the large relevant equation could be typed out and solved.

Research virtual machines (RVM) are useful for computationally expensive operations. While they weren't needed when using the GPR method, my project partner, Daniel Clark, found it necessary to use RVMs to solve his optimization formulations. For more additional information about RVMs, please refer to the University of Auckland's research hub [25].

3.3.1 ccandu

A key tool for producing conditional GPR models is called ccandu, pronounced as "can do", which is an in-house custom library built by the Modelling Uncertainty and Data Group at the University of Auckland [26]. It's a highly specialized tool that excels at utilizing conditional random variables by combining and modifying them. Using this library allows for custom mean and covariance functions, as well as custom kernels, to be set based on relevant input. Corresponding posterior distributions that satisfy given summative constraints could be readily drawn due to the conditional capabilities of this tool. Using ccandu offers a flexible and creative approach to implementing the GPR model, as there is plenty of freedom to define aspects of the GPR model itself. As detailed later on in the report, ccandu replaces Scikit-Learn as the library to use for generating GPR models due to the reasons described and overall better performance.

Chapter 4

Results

The implementation of GPR on problems of varying difficulty is outlined, along with the quality of their predictions. Synthetic data is initially used to fit and test GPR models. Initially, a simple problem is solved to generate a proof of concept. Afterwards, more challenging problems are solved where assumptions are relaxed further. Later on, real-life data is used for the GPR models to observe the performance of these current methods on problems that are likely to be encountered by geothermal operators. Finally, it will be determined whether there's a substantial difference between the GPR and mathematical formulation method and which ones, if any, perform better for the varying problems.

4.1 Independent GPR Processes

The simplest case of implementing a GPR model for this problem is considering only two wells exist in the geothermal field, and these two wells are always on. Scikit-Learn's implementation of a GPR model requires an appropriate kernel, interchangeably referred to as a covariance matrix, to be specified. The earliest implementation of this GPR model uses a RationalQuadratic kernel. Mathematically, this kernel is expressed as

$$k(x_i, x_j) = \left(1 + \frac{d(x_i, x_j)^2}{2\alpha l^2}\right)^{-\alpha} \quad (4.1.1)$$

where α is the scale mixture parameter, l is the length scale, and $d(x_i, x_j)$ is the Euclidean distance between x_i and x_j . Using this, two GPR models are fitted for the two wells using $l_{\text{bounds}} = (0.1, 100)$ and $\alpha = 10^{-10}$. These GPR models are fitted independently to each other and the total steam flow. Figure 7 displays the results of these two separate GPR models.

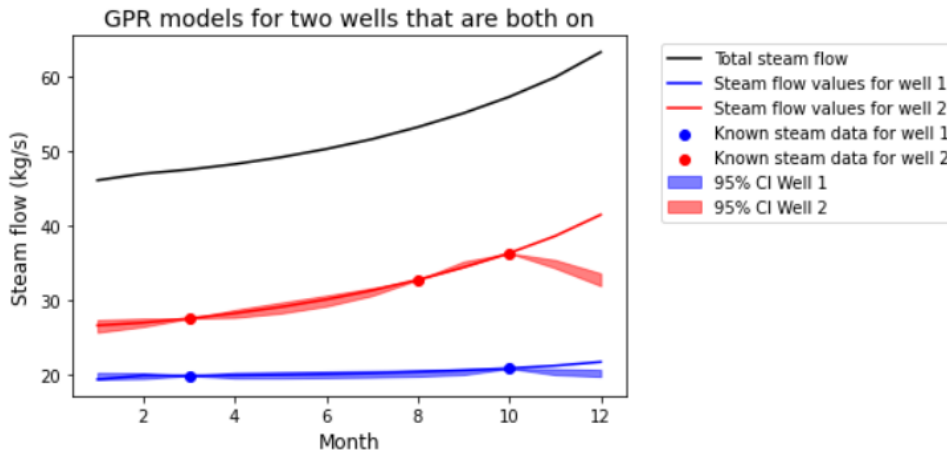


Figure 7: GPR models for two wells that are always on.

The GPR models for both wells capture the true steam flow values within their uncertainty intervals up to month 10. After this, the uncertainty bands fail to capture the steam flow values for months 11 and 12 as the predicted steam flow for both wells drops sharply when it should instead rise. This suggests the GPR model struggles with the extrapolation of steam flow values, which is a very common issue with machine learning algorithms. While the uncertainty intervals did capture the true steam flow values up until month 10, they were far too narrow and indicated a high level of confidence in the model created. With such sparse data and numerous potential applications, widening the uncertainty intervals would create a more flexible model that would better capture the broad range of possible values for missing steam flows. Adjusting the kernel by weakening the correlations between recorded steam flows, as well as introducing a scaling factor, would adjust the width of the produced uncertainty intervals. A Matern kernel was also used to see whether this generalized kernel was potentially better suited than the RationalQuadratic kernel. The Matern kernel is given below and can be expressed as

$$k(x_i, x_j) = \frac{1}{\Gamma(\nu)2^{\nu-1}} \left(\frac{\sqrt{2\nu}}{l} d(x_i, x_j) \right)^\nu K_\nu \left(\frac{\sqrt{2\nu}}{l} d(x_i, x_j) \right) \quad (4.1.2)$$

where ν controls the smoothness of the learned function, l is the length scale, $d(x_i, x_j)$ is the Euclidean distance between x_i and x_j , K_ν is the modified Bessel function, and $\Gamma(\nu)$ is the gamma function. The refitted GPR models using the RationalQuadratic kernel used $l_{\text{bounds}} = (0.1, 10)$ and $\alpha = 10^{-10}$. The GPR models using the Matern kernel also used $l_{\text{bounds}} = (0.1, 10)$ and $\alpha = 10^{-10}$ with a new parameter, ν , being set to 1.5. Figure 8 visually illustrates the differences between using the RationalQuadratic and Matern kernel to solve the same problem. Both the RationalQuadratic and Matern kernel create GPR models that have their uncertainty intervals capture the true steam flow. While there wasn't a drastic difference between the GPR models fitted for well 2 when using a RationalQuadratic or Matern kernel, there were large differences observed for well 1. Taking the 25th and 75th quantile of variance for well 1 predictions across GPR models fitted gives $\text{Quantile}_{\text{RationalQuadratic}} = (0.44, 0.56)$ and $\text{Quantile}_{\text{Matern}} = (4.39, 17.95)$. Repeating this for well 2 gives $\text{Quantile}_{\text{RationalQuadratic}} = (1.48, 10.93)$ and $\text{Quantile}_{\text{Matern}} = (0.95, 11.32)$. There is little difference in the variance quantiles when using either a RationalQuadratic or Matern kernel for well 2. However, there is a large difference when using a Matern kernel over a RationalQuadratic kernel as the well 1 variance quantile has drastically increased. This suggests that the Matern kernel produces GPR models with larger uncertainty intervals, which is desirable because the model needs to capture various possible steam flows for missing data through its uncertainty intervals. As such, even though the Matern kernel performed worse, its ability to create wider uncertainty intervals offers more flexibility and confidence when dealing with extremely sparse data and, therefore, is the kernel used for future GPR implementations using Scikit-Learn.

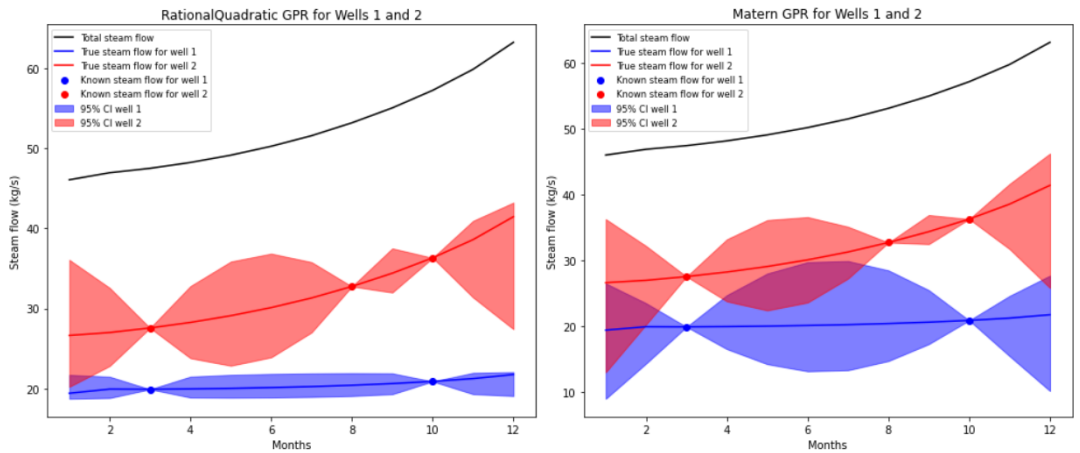


Figure 8: (Left): GPR models fitted using a RationalQuadratic kernel.
(Right): GPR models fitted using a Matern kernel.

The geothermal operator or whoever has interest could draw realizations for steam flow for both wells across the 12-month period. These drawn realizations could then be converted back to mass flow by dividing through by dryness and solving for the missing mass flow values. A key constraint that needs to be satisfied is that the sum of steam flows across the two wells at a given time point would sum exactly to the total steam flow at the same time point. Equation (4.1.3) represents this constraint for this current problem, while equation (4.1.4) represents this in a general case.

$$\text{Total} = S_{\text{Well}_1} + S_{\text{Well}_2} \quad (4.1.3)$$

$$\text{Total} = \sum_{i=1}^n S_{\text{Well}_i} \quad (4.1.4)$$

where S_{Well_i} is the steam flow for the i th well. Conditioning is a logical solution to enforcing this constraint and is rightfully discussed in the subsequent section.

4.2 GPR Processes with Conditioning

Using (4.1.4) and (4.1.3), the previous problem where two wells exist in the geothermal field and are assumed to always be on can be resolved. Scikit-Learn doesn't have the ability to formulate GPR models under a given constraint, so the conditioning process is rather crude. Instead, a semi-manual approach is implemented when solving this conditioning problem.

Consider equation (4.1.3) for the two-well example. It can be rearranged such that the subject is now a particular well:

$$S_{\text{Well}_i} = \text{Total} - S_{\text{Well}_j} \quad (4.2.1)$$

where S_{Well_i} is the steam flow for the i th well and S_{Well_j} is the steam flow for the j th well at a given time. In the two-well case, $i, j \in \{1, 2\}$, where $i \neq j$.

Therefore, the GPR model can be refitted as before but for time points where the steam flow is known for one well but not the other, the exact steam flow for the unknown well can be deduced at the time point. This essentially clips the uncertainty interval for both GPR models, allowing for GPR models to still succeed with even less data. Once fitted, the mean predictions for steam flows are summed together to find the total mean steam flow predicted across both wells. The difference between this total and the true total steam flow is then calculated and both GPR model mean predictions are adjusted by this amount. Mathematically, this is given by:

$$\text{MP}_{\text{Well}_i} = \frac{\text{TS} - \text{MP}_{\text{Sum}}}{2} + \text{MP}_{\text{Well}_i} \quad (4.2.2)$$

where $\text{MP}_{\text{Well}_i}$ is the mean prediction for the i th well, TS is the true total steam flow, and MP_{Sum} is the sum of mean predictions. Dividing by two in the case ensures that, since there are two wells in this case, they both are scaled the same way. To be complete, the generalized expression can be given as:

$$\text{MP}_{\text{Well}_i} = \frac{\text{TS} - \text{MP}_{\text{Sum}}}{n} + \text{MP}_{\text{Well}_i} \quad (4.2.3)$$

where everything is the same as equation (4.2.2) except the division term is n where n is the number of GPR models fitted. The newly fitted GPR model using this conditioning method is shown in figure 9. The conditioned GPR models show immense improvement

over the previous GPR models as now extra information about wells is incorporated during fitting, indicated by the clipped uncertainty intervals for both wells at time points where at least one well's steam flow is known, and the trends for both models have been corrected.

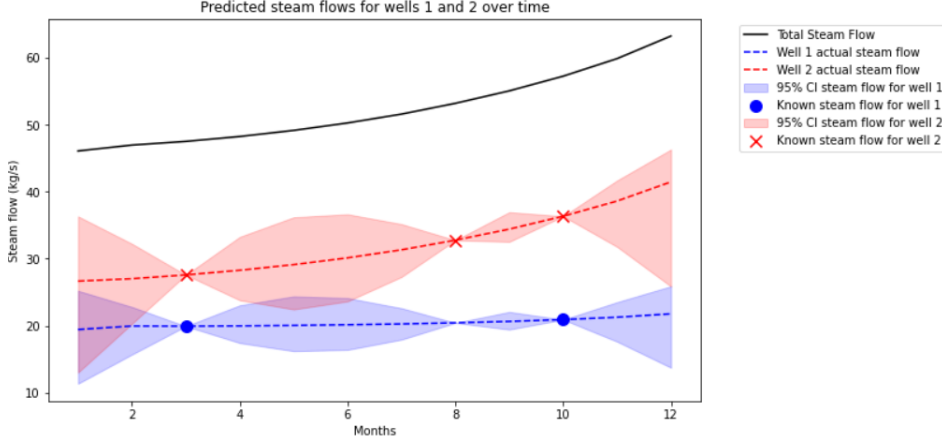


Figure 9: GPR model with conditioning.

Realizations of the steam flows across the entire time period for both wells can be drawn. This is done in a rather crude yet sensible way where a realization is drawn for well 1 across the time period, and then the subsequent realization is drawn for well 2 while enforcing the constraint given by (4.1.3). The realizations drawn are for steam flow, meaning they must be divided through by the corresponding dryness to obtain the mass flow. Figure 10 demonstrates a single steam flow realization being drawn across the 12-month period for both wells 1 and 2 with the equivalent mass flow realizations.

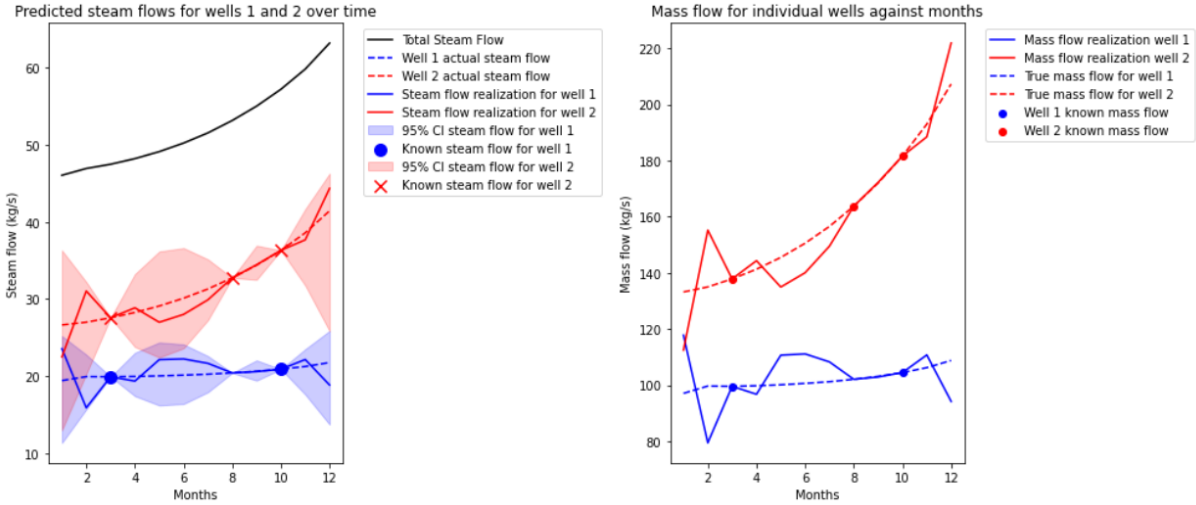


Figure 10: (Left): A single steam flow realization across 12 months for wells 1 and 2. (Right): The corresponding mass flow realization for wells 1 and 2.

From figure 10, the left plot clearly demonstrates that the realizations drawn are bounded within the uncertainty intervals created. When the steam flow realizations are converted back to mass flow, which is given in the right plot, there is a mix of result quality. When the steam flow for either well is known, the corresponding mass flow prediction matches the true mass flow exactly. Otherwise, the mass flow realizations range in quality with $e = (0.22, 20.71)$ where e is the residual between the mass flow realization and true mass flow.

These results prove that conditioning in GPR models works and successfully enforces (4.1.4). However, this method of conditioning using Scikit-Learn is undoubtedly considered crude and has become extremely challenging to scale for more than 2 GPR processes.

The ccandu library, developed by the University of Auckland modelling team, is another potential package that could do this conditioning process. As the package is specifically designed for Gaussian processes and conditioning methods, it's likely the results would yield more potential.

4.2.1 Improved Conditioning Using the ccandu Library

The same problem is solved using the ccandu library where there are two wells in the geothermal field and they both remain on for the entire time period. Setting up the conditioning problem is a more manual process than in Scikit-Learn but offers more control and flexibility than Scikit-Learn. A prior mean and covariance matrix is defined for each individual GPR model that needs to be fitted. Then, the prior model is created by using the total steam flow and the previously fit priors. Next, the posterior distribution is defined by using the data observations given. Combining the prior and posterior distribution allows for posterior samples to be drawn for each GPR model fitted. The sum of these posterior samples across the GPR models fitted naturally sum to the total steam flow, enforcing constraint (4.1.4). Figure 11 demonstrates using ccandu to solve this two well that's always on problem.

Figure 11 demonstrates more robust samples being drawn throughout the entire time period, except for the extrapolation period. Compared to ?? the intervals are more narrow and hence are more accurate up until the extrapolation period. The extrapolation period sees samples drawn with a more narrow interval but behave oddly as well 1 predictions increase when they are expected to remain rather flat and well 2 predictions dip when they were expected to continue rising.

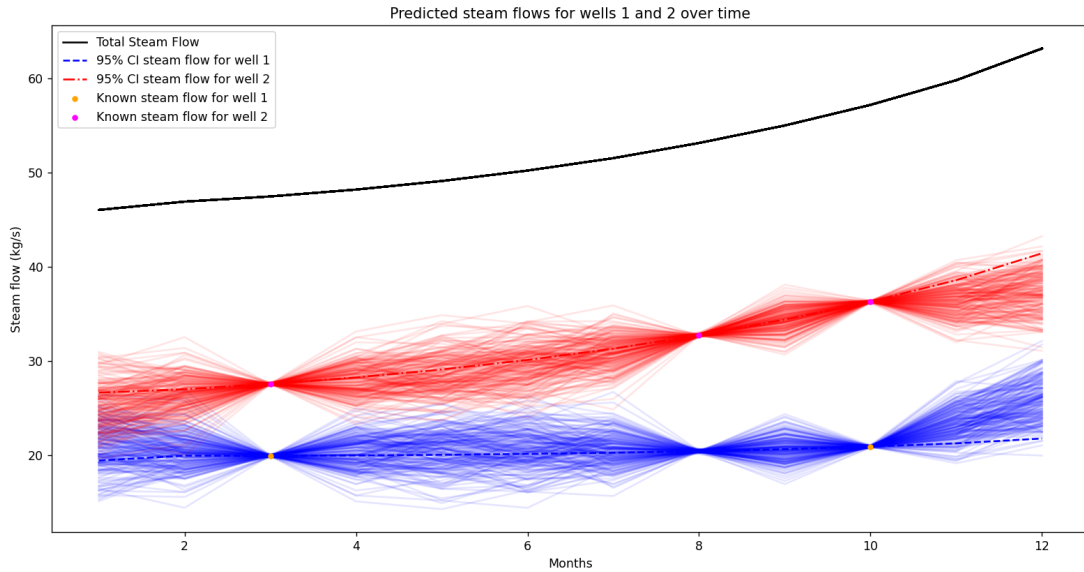


Figure 11: GPR model with conditioning and realizations for both wells using ccandu.

As seen previously in ?? and ??, relevant posterior samples for steam flow can be drawn and readily transformed back to mass flow. Figure 12 displays an example of using ccandu to generate a steam flow realization for both wells across the time period.

Converting this steam flow realization shown in figure 12 gives mass flow realizations shown in figure 13.

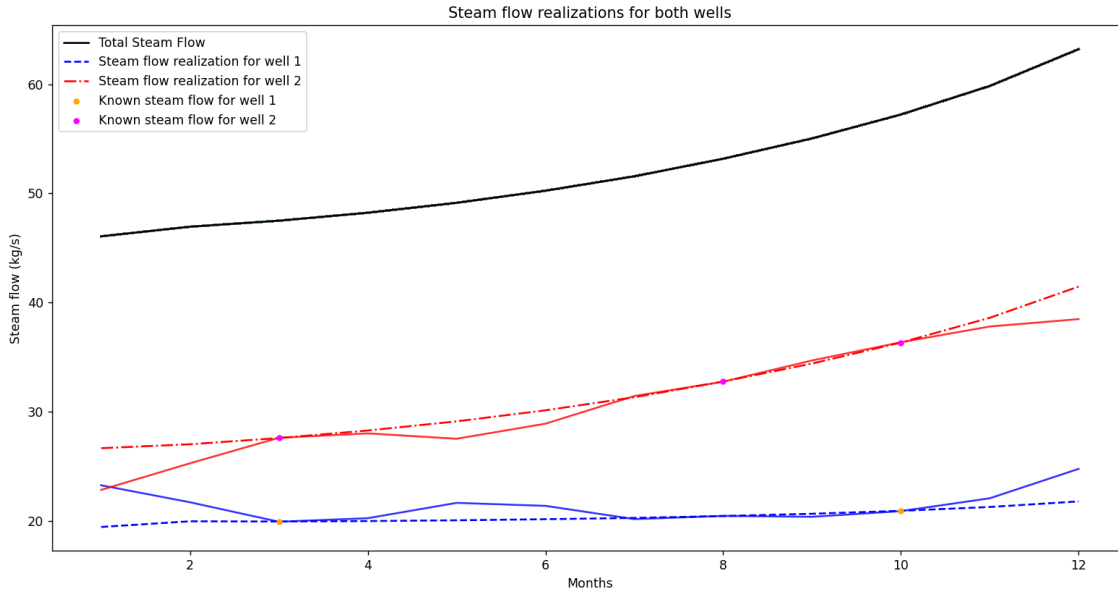


Figure 12: Steam flow realizations for both wells using ccandu.

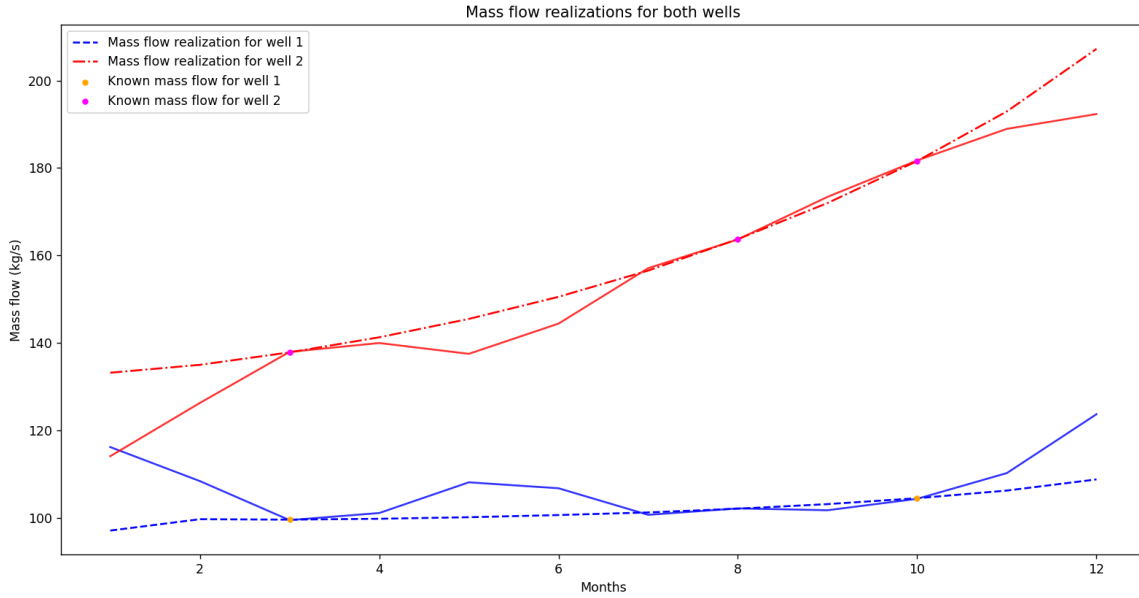


Figure 13: Mass flow realizations for both wells using ccandu.

It has been demonstrated that GPR models can be fitted using both the ccandu and Scikit-Learn library such that steam flow realizations, and consequently mass flow realizations, can be drawn from them. A proof of concept has been generated using both the ccandu and Scikit-Learn library. Relaxing further assumptions allows for the GPR model to tackle more complex problems, and is discussed in more detail in the subsequent section.

4.3 GPR with Further Relaxed Assumptions

Previously, the problem solved only considered two wells in the geothermal field and assumed these wells were always on. This allowed for a proof of concept to be created but doesn't represent a real-life problem very well. Relaxing further assumptions brings on the challenge of solving problems where multiple wells exist in the geothermal field, and these wells can be on or off. This section investigates how to solve these problems using the ccandu library, as the ccandu library proved to be a robust way of enforcing realization sums and drawing valid posterior realizations.

Consider an expansion of the previous problem where there are 5 wells in the geothermal field and they turn on at different times. The same process as before can be applied where two GPR models are fitted. This works by selecting a target well and then grouping all the remaining wells into a lumped well. Mathematically, it's exactly the same as (4.1.4), but for this problem can be written as

$$\text{Total} = S_{\text{Well}_i} + S_{\text{Others}} \quad (4.3.1)$$

where S_{Others} is the steam flow across all the wells excluding the i th well. S_{Others} can be written as

$$S_{\text{Others}} = \sum_{\substack{j=1 \\ j \neq i}}^n S_{\text{well}_j} \quad (4.3.2)$$

which finds the sum of steam flow across all the wells in the geothermal field except the i th one, as this is our target. Suppose that in the 5 well geothermal field there is interest in determining the steam flow, and thus mass flow, across well 1, then applying the processes described earlier results in figure 14. Figure 14 demonstrates the ability to generate steam flow realizations that are fairly accurate between known periods of steam flow recordings. The prediction intervals generated for the well 1 and other well steam flows up to and including month 30 comfortably capture the true steam flow values. As alluded to before, there is a struggle to capture accurate behaviour after the last recorded data point and it's evident here with both GPR models failing to have the prediction intervals capture the true steam flow.

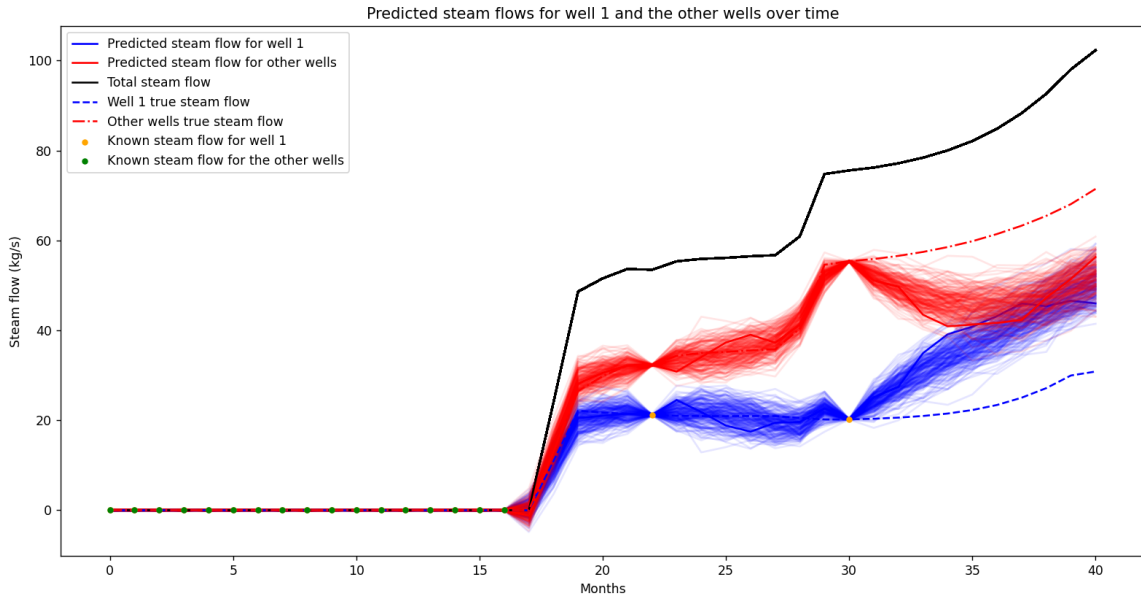


Figure 14: GPR model for multiple wells that can be on and off where well 1 is the target well and the other wells are lumped together.

A geothermal operator could use this setup very effectively to interpolate steam flow values for the relevant geothermal well. By giving a target well, two GPR models would be fitted and the GPR model fit for the target well would generate a reasonable steam flow prediction, assuming no extrapolated predictions are used, that could easily be converted to the mass flow using dryness. This method is suited for inferring the detailed history of a particular well. However, it's undeniably limited when trying to infer the history of many wells, which is a common problem. This approach only helps gain insight into one well as the lumped well is essentially meaningless. Fortunately, this approach can be extended through the introduction of fitting GPR models for each individual well in the geothermal field.

4.3.1 Fitting GPR Models to Each Well

Up until now, every problem has been solved by fitting two GPR models. If there are two wells in the geothermal field then a GPR model is fitted for each well otherwise all non-target wells are simply lumped into a lumped well and a GPR model is fitted to this lumped well and target well. While challenging, a GPR model can be fit for each well in the geothermal field. This adds robustness as each well is treated separately and the geothermal operator can obtain predictions for all wells simultaneously.

Solving the same 5-well problem but now fitting a GPR model for each individual well results in figure 15 where 5 separate GPR models have been fitted to each of these 5 wells. There are mixed results using this method as some uncertainty intervals produced successfully capture the true steam flow behaviour within but others don't. Specific examples of this from figure 15 are seen from wells 3 and 5. The well 3 uncertainty intervals for steam flow between months 18 and 21 fail to capture the true steam flow values within this interval. Likewise, the uncertainty intervals for well 5 between months 26 and 29 also fail to capture the true steam flow values within this interval.

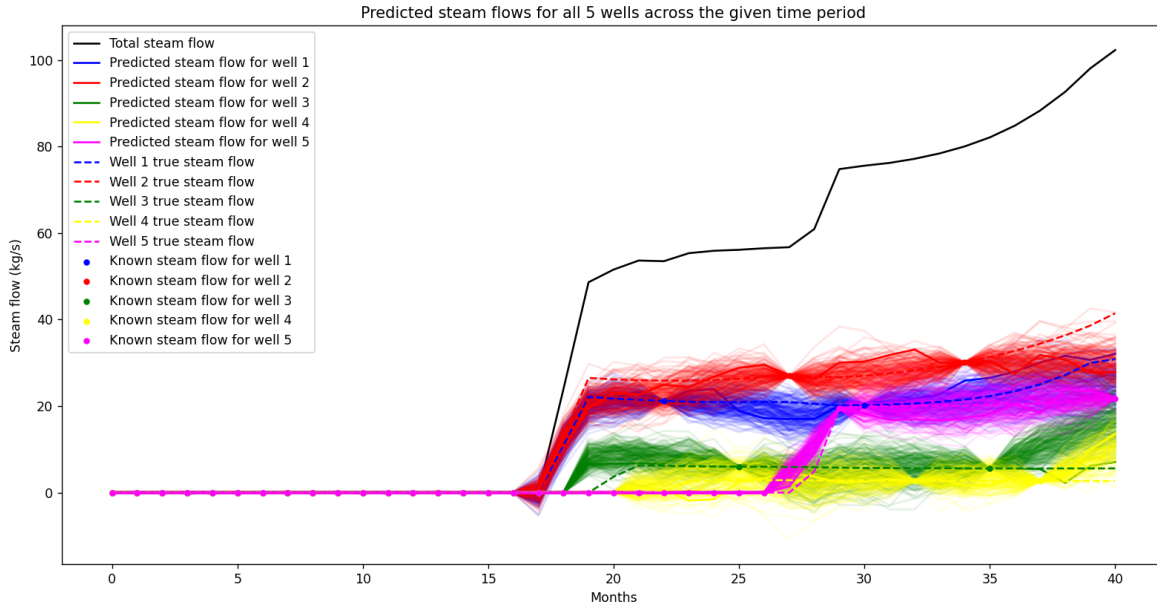


Figure 15: GPR models fitted for 5 wells that can be turned on and off.

This section has demonstrated an effective way to solve the sparse data problem using individual GPR models fitted for each well under given constraints. While the results have been mixed, there are elements to this approach that are favourable over the lumped GPR model approach previously described, but this will be discussed in detail later. Unexpectedly, the GPR models for all 5 wells performed poorly with extrapolation, showing little improvement over the lumped GPR model approach.

4.4 Using GPR on Real-Life Data

Real-life data is much more challenging to work with compared to synthetic data. Often, data collected from geothermal fields are even more sparse than synthetic data, have the potential to be erroneous, and are supplied in crude formats that require prior processing. Contact Energy [27] has supplied a broad range of geothermal data collected from their various geothermal fields. This supplied data contains the recorded total mass flow as opposed to the total steam flow, meaning there's no need to convert steam flow back to mass flow. Individual mass flow measurements are also sometimes reviewed by reservoir engineers to ensure their validity and to check for potential issues.

GPR models were used to generate predictions for missing mass flows for the collected data supplied by Contact Energy. Data from a separator containing 6 wells spanning over

1000 days was used as a subset of the total collected data. In this run, individual mass flow measurements were weighted and thus treated equally regardless of whether they were measured by a geothermal reservoir engineer or not. Applying the methods seen in 4.3.1 to this collected data results in figure 16 below. The first observation from figure 16 is the enhanced complexity of the separator behaviour. The total mass flow experiences sudden spikes and drops and has no clear trend, a clear contrast to what has been observed using synthetic data. The time period to model for is also significantly higher than what was done for before with 1000 points now being modeled, instead of 40 using the synthetic data. The complex interactions of the wells within this separator lead to a mixture of performance across different wells at different times. One of the challenges with validating the accuracy of the GPR predictions is that the expert estimate, indicated by the dashed lines in figure 16, is not necessarily the true mass flow for the individual wells. Rather, it's an estimate made by an expert in the geothermal space. As such, the GPR model performance is validated against results that aren't guaranteed to be right resulting, albeit unlikely, in the case where both the GPR predictions and expert estimates are far off from each other. This is especially evident for well 2 where the GPR predictions fail to capture the expert's estimates for a significant portion of the 1000 day period. The remaining wells fare much better where the GPR predictions capture the expert's estimates for a reasonable number of days. One pleasing behaviour observed through wells 4, 5, and 6 is where the wells switch on and off. Well 4 only switches on towards the end, around day 900, yet doesn't exhibit any unusual behavior prior to this with the GPR predictions being approximately zero for day 900 and earlier. Well 5 and 6 remains off the entire 1000 days and the GPR models only predict a mass flow rate of approximately 0 across this time period.

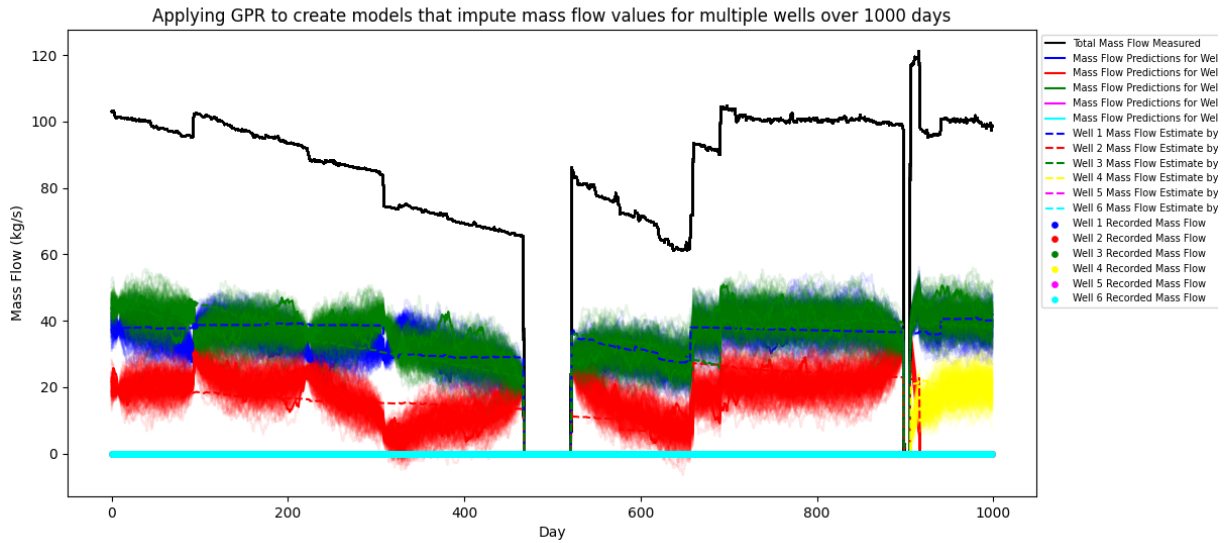


Figure 16: GPR models fitted on real-life data for a separator containing 6 wells over a period of 1000 days.

To facilitate confidence in the GPR model, a section of the time period is clipped to demonstrate the uncertainty interval narrowing through the introduction of measured mass flows for individual wells. Figure 17 below highlights a key interval within the 1000 days where there are mass flow values for wells 2 and 3. The uncertainty interval becomes clipped here meaning the GPR models are considering these data points when generating predictions. Consequently, there is promising evidence of the GPR models performing to their expectations.

Quantifying the corresponding results for the real-life application is crucial to understanding the quality of predictions made and potential pitfalls. Firstly, confidence intervals for the GPR predictions made across all wells are given below in figure 18. The confidence intervals produced reveal some interesting results. There are periods where the GPR

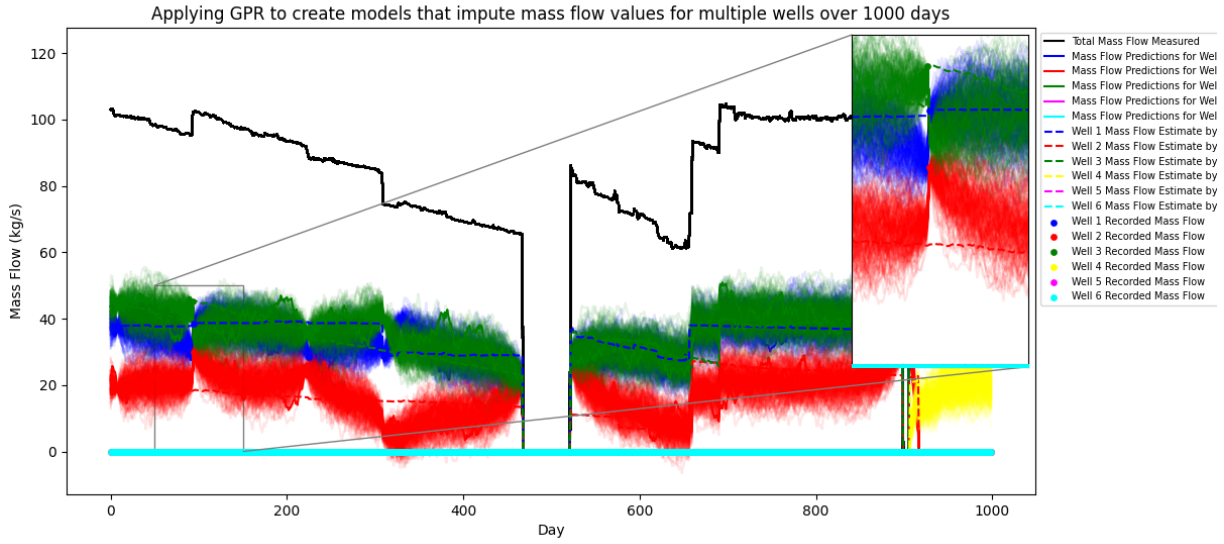


Figure 17: GPR models fitted on real-life data for a separator containing 6 wells over a period of 1000 days with a zoomed-in section.

model confidence intervals completely miss the expert's estimate and other times aligns rather well with the expert's estimate. While the GPR models uncertainty intervals all capture the recorded individual mass flow values, the expert's estimates occasionally miss the recorded mass flow values with them not being remotely close. For example, at around day 100 a mass flow of around 30 kg/s was recorded for well 2. While the GPR model confidence interval for well 2 captured this mass flow value, the expert's estimate was just under 20 kg/s. Further discussion into the potential implications of this are detailed in the discussion section.

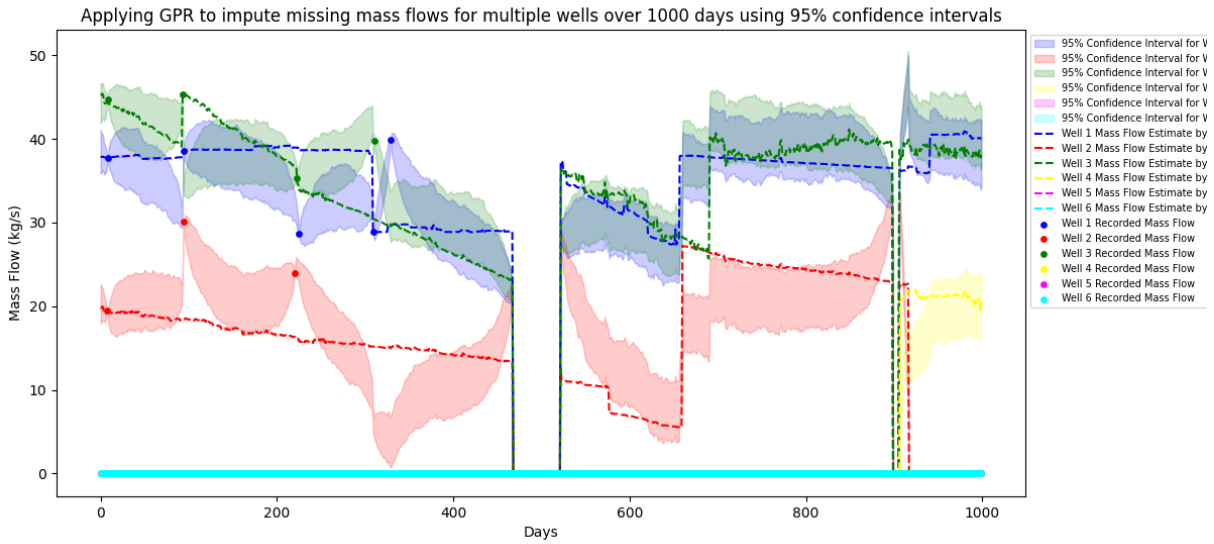


Figure 18: GPR models fitted on real-life data for a separator containing 6 wells over a period of 1000 days with relevant 95% CI

Using the confidence intervals obtained, the expert's estimates, and the measured individual mass flows allows for further insight to be gained from applying GPR to solving this real-life problem. Table 4.1 shows the relevant performance metrics for the GPR model on this real-life application where the minimum difference is the smallest possible difference between the upper and lower bound for the relevant 95% confidence interval, maximum difference is the same except it's the largest possible difference, and the average RSS

score is taking the median value of the confidence intervals and comparing this to the expert's estimate by taking the square of this difference, summing it, and then dividing through by the number of discrete time points. Table 4.1 explains the results seen in figures 16, 17, and 18 concisely. The average RSS score for wells 5 and 6 are essentially zero and, since both these wells are off, indicates the GPR model is able to handle the on and off behaviour wells exhibit. Wells 1, 2, and 3 show a high level of variability in their performance metrics. While their minimum and maximum difference remain roughly the same as each other, the average RSS score of well 2 is extremely high with it being just over 23. This is unsurprising as the GPR model predictions for well 2 were consistently missing the expert's estimates.

Table 4.1: Differences and Average RSS Scores for Wells 1-6

Well	Minimum Difference	Maximum Difference	Average RSS Score
Well 1	0.1458	8.4414	11.7390
Well 2	0.1445	8.1305	23.0137
Well 3	0.1477	8.2815	14.7462
Well 4	0.1469	7.6847	1.9335
Well 5	0.1470	0.2152	0.0002
Well 6	0.1511	0.2168	0.0002

4.5 GPR Versus Mathematical Formulation

Chapter 5

Discussion

5.1 Findings

5.2 Extrapolation

5.3 Further Work

Chapter 6

Conclusion

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